A certified median threshold for six-variable random 2-SAT

Nefedov Sergey
Moscow State University, Mathematics Department

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Version 3: reproducibility repository

Abstract

We evaluate the number of satisfiable thirteen-clause sets on six labeled Boolean variables in the distinct-clause model. The exact count is 2 381 437 338 960, giving probability $9\,922\,655\,579/21\,528\,597\,615 < 1/2$. Together with the known twelve-clause count and monotonicity, this proves that thirteen is the first strict half-probability crossing. The evaluation uses the existing enumeration theorem of Dovgal, de Panafieu and Ravelomanana and is independently reproduced by a direct assignment-mask dynamic program. We supply integer-polynomial certificates, source code and executed verification logs. The contribution is a certified finite evaluation of existing theory; no new general enumeration theorem is claimed. The exact certificate arithmetic, the finite clause model and probability monotonicity have also been checked in Lean 4. The identification of the certificate coefficients with the satisfiable-formula counts is not yet formalized, so the Lean verification is partial.

1 Model and result

Let $\mathcal{C}_n$ contain the four signed disjunctions on every unordered pair of distinct variables from $x_0, \dots, x_{n-1}$. Thus $|\mathcal{C}_n| = 4\binom{n}{2} = 2n(n-1)$. A formula is an unordered subset of $\mathcal{C}_n$. Variables are labeled; clauses cannot repeat. Write

$$a_{n,m} = \#\{F \subseteq \mathcal{C}_n : |F| = m,\ F \text{ satisfiable}\}, \qquad P_{n,m} = \frac{a_{n,m}}{\binom{2n(n-1)}{m}}.$$

The finite request P2588 of TheoremDB asks for the first m with $P_{6,m} < 1/2$.

Theorem 1. *In this model the first such value is $m = 13$. More precisely,*

$$a_{6,12} = 805\,717\,285\,720, \qquad a_{6,13} = 2\,381\,437\,338\,960,$$

and

$$P_{6,13} = \frac{9\,922\,655\,579}{21\,528\,597\,615} = 0.46090580336205517\ldots.$$

The twelve-clause value is in the published table of [1]. We reproduced it before evaluating the next coefficient. The exact integer comparisons are

m	$\binom{60}{m}$	$2a_{6,m} - \binom{60}{m}$
12	1 399 358 844 975	+212 075 726 465
13	5 166 863 427 600	−403 988 749 680

Lemma 2. *For a fixed finite clause universe of size C, satisfiability probabilities of uniform clause sets satisfy $P_{m+1} \le P_m$ for $0 \le m < C$.*

Proof. Choose a uniform $(m+1)$-set and delete a uniformly chosen clause. A fixed m-set has $C-m$ extensions, and its probability under deletion is

$$\frac{C-m}{(m+1)\binom{C}{m+1}} = \frac{1}{\binom{C}{m}}.$$

The resulting m-set is therefore uniform. Deletion preserves satisfiability, so the satisfiability indicator cannot decrease under this coupling. Taking expectations proves the claim. □

Consequently the two integer comparisons prove the claimed first crossing, once the coefficient evaluations below are certified.

2 Exact coefficient evaluation

Set $q = 1 + w$ and use exponential generating functions in z:

$$G(z,w) = \sum_{r\geq 0} q^{\binom{r}{2}} \frac{z^r}{r!}, \qquad H = G^{-1}, \qquad E = G \odot_z H, \qquad A = \sqrt{E},$$

where $A(0,w) = 1$ and $\odot_z$ is the exponential Hadamard product. Theorem 4.7 of [1] gives, writing $A_r = r!\,[z^r]\,A$,

$$a_{n,m} = [w^m]\ 2^n \sum_{k=0}^{n} \binom{n}{k} q^{2(n-1)k-\binom{k}{2}} A_{n-k}(w). \tag{1}$$

This is the source enumeration theorem, not a new formula of this note. The normalization distinguishes labeled variables, unordered clause sets and exponential coefficients explicitly.

For $H_r = r!\,[z^r]\,H$ and $E_r = q^{\binom{r}{2}} H_r$, formal inversion and square-root extraction give

$$H_0 = 1, \qquad H_r = -\sum_{j=1}^{r} \binom{r}{j} q^{\binom{j}{2}} H_{r-j}, \tag{2}$$

$$A_0 = 1, \qquad 2A_r = E_r - \sum_{j=1}^{r-1} \binom{r}{j} A_j A_{r-j}. \tag{3}$$

The primary implementation evaluates these identities as full polynomials in w over $\mathbb{Q}$, with exact rational arithmetic. Coefficients above z^n cannot affect the requested result: each recurrence for degree r uses only degrees at most r. All w-degrees were retained, so no truncation in the clause variable is involved. Full rows were computed for $n \leq 7$.

2.1 An integer certificate

A separately written evaluator uses ordinary powers of q and the scaled coefficients $R_r = 2^r A_r$. Then

$$R_0 = 1, \qquad 2R_r = 2^r E_r - \sum_{j=1}^{r-1} \binom{r}{j} R_j R_{r-j}. \tag{4}$$

The final polynomial and its coefficient transform are

$$S_n(q) = \sum_{k=0}^{n} 2^k \binom{n}{k} q^{2(n-1)k-\binom{k}{2}} R_{n-k}(q), \tag{5}$$

$$a_{n,m} = \sum_{j\geq m} \left[q^j\right] S_n(q) \binom{j}{m}. \tag{6}$$

The stored certificate contains the integer coefficient arrays H_r, E_r, R_r, S_r. The stand-alone checker verifies $GH = 1$, the Hadamard identity and $R(z)^2 = E(2z)$, then checks (1)–(6) and all saved coefficient rows. It imports neither evaluator and needs no symbolic algebra package. Uniqueness of the formal inverse and the square root with constant coefficient 1 justifies the finite identity certificate.

3 Independent direct enumeration

The second counting method uses no generating functions. Encode the 64 truth assignments by bits of a 64-bit unsigned integer. For a clause c, let $M(c)$ have a bit precisely at assignments satisfying c. Each clause mask has 48 bits. The solution set of a formula is exactly $\bigcap_{c\in F} M(c)$.

After the first i clauses, let $D_i(M, k)$ count k-element subsets with exact solution mask M. Initially $D_0(\mathrm{ALL}, 0) = 1$. Every clause produces the two transitions

$$(M, k) \longrightarrow (M, k), \qquad (M, k) \longrightarrow (M \,\&\, M(c), k + 1).$$

The two alternatives partition all subsets by absence or presence of the new clause. Induction therefore proves that the DP counts each labeled clause set exactly once. Equal masks are merged by addition of multiplicities, not by identification of formulas. An empty mask can be discarded permanently; degrees above 13 can also be discarded.

In-place evaluation. To reduce memory, take a snapshot of the existing state nodes before each clause and process them in increasing mask population count. A changed mask is a proper subset and has strictly smaller population count. Its node is therefore already processed or is new and absent from the snapshot. Hence no earlier transition can alter the source polynomial before it is read. If the mask is fixed, update degrees in decreasing order, multiplying its polynomial by $1 + w$. This implements the same recurrence as the simple two-buffer reference program.

For $k \le 13$, every count is bounded by $\binom{60}{13} < 2^{64}$, so the counting arithmetic cannot overflow. The state checksums intentionally use arithmetic modulo 2^{64} and play no role in the counts.

Executed output. All 60 steps completed. There were 24 444 343 nonempty final masks; the last step was recorded at 117.657 seconds. All fourteen counts, $0 \le m \le 13$, agreed with the primary calculation. The final two entries were exactly the values in Theorem 1. The run logs retain a deterministic order-independent 64-bit checksum for every state table. SHA-256 digests cover distributed source, certificates, logs and outputs. The large state tables are regenerated on demand and are not distributed; the 64-bit checksums are not cryptographic certificates.

3.1 Additional executed checks

- Every one of the $2^{24} = 16\,777\,216$ clause sets at $n = 4$ was enumerated separately. The complete row agrees in all 25 degrees. A simple two-buffer Python DP also agrees for every degree at $n \le 4$.
- For all 4096 formulas at $n = 3$, truth-table satisfiability agrees with the implication-graph SCC criterion. A further fixed-seed test checks 1220 formulas at $n = 6$. These tests are not probability estimates.
- Both unmodified coefficient functions from the pinned upstream code were executed at $(6, 12)$ and $(6, 13)$ and returned the same counts.
- The published totals for $n \le 7$ were reproduced. In particular,

$$\sum_m a_{6,m} = 2\,034\,602\,766\,692\,687.$$

  Its complement is 1 150 886 901 840 154 289, and the two sum to 2^{60}.

4 Reproducibility and scope

The intended repository URL for these materials is

https://github.com/Sergpreneur/P2588-2SAT-Threshold.

Public upload is pending for this version; the accompanying archive contains the complete materials prepared for that repository. The package includes both polynomial evaluators, the independent C++ mask DP, the exhaustive checker, full data, certificates, logs, version information and the Russian report. From the package root:

```
python3 verify/verify_count.py
python3 -m unittest discover -s tests -v
python3 verify/verify_count.py --rerun-dp
```

The first command verifies the compact certificate and saved outputs; the last also recompiles and reexecutes the independent large DP. The recorded environment used Python 3.12.14 and g++ 13.3.0 with `-O3 -std=c++17`. SymPy 1.14.0 was used only to run upstream code. The main programs and the compact checker use the Python standard library. The C++ DP requires several GiB of memory. These computational checks are complemented by the partial Lean verification described below.

5 Lean 4 verification and its scope

The accompanying formal project was compiled with Lean 4.19.0 and mathlib v4.19.0. The pinned mathlib revision is

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The standard `lake build` completed successfully; the build and axiom reports are included in the `lean/` directory, together with the formal sources.

The arithmetic modules check the finite integer-polynomial identities for H_r, E_r, R_r through degree six, the assembly of S_6, the binomial transforms at twelve and thirteen clauses, both binomial denominators, the exact signed gaps, and the reduced probability. They also check every strict above-half comparison for the coefficients at $0 \le m < 13$. These modules depend only on Lean's standard library and use ordinary `decide`, whose proof terms are checked by the kernel. The reported arithmetic theorems have no axiom dependencies; neither `native_decide` nor an external computational oracle is used.

The separate model module defines a clause by a variable pair $i < j$ and two Boolean signs, represents formulas as finite sets, and defines satisfiability by an existential truth assignment. Lean proves that the clause universe has cardinality sixty and that there are $\binom{60}{m}$ formulas of size m. It also proves deletion preserves satisfiability and that the corresponding probability is nonincreasing. This last proof uses the local Lubell–Yamamoto–Meshalkin inequality from mathlib. Its axiom dependencies are the standard `propext`, `Classical.choice` and `Quot.sound`. There are no admitted proof terms or new enumeration axioms.

The remaining formal gap is explicit. Writing `satCount` for the count of satisfiable formulas in the Lean model, the final theorem `threshold13_of_enumeration_bridge` assumes

$$\texttt{satCount}(12) = \left[w^{12}\right] S_6(1+w),$$
$$\texttt{satCount}(13) = \left[w^{13}\right] S_6(1+w).$$

Under these hypotheses it proves that thirteen is the first strict crossing. The hypotheses are not proved inside Lean: the specialization of the source enumeration theorem, or a verified counting

procedure connecting the coefficient certificate to this finite model, remains to be formalized. Thus the numerical and monotonicity parts are kernel-checked, while the complete counting argument and the unconditional threshold theorem are not yet fully formalized in Lean. The mathematical result rests on [1] and the independent executed enumeration described above.

Literature and priority. The general counting problem is already solved by [1]. The latest displayed arXiv version was v2, and the inspected GitLab master head remained commit `346079fa8606ebd48f4557bcef7bc1f73a2aed4d`. Its notebook displays the row only through $m = 12$, but also computes totals by summing all degrees. Thus it already evaluated $a_{6,13}$ internally. A bounded search found no separate printed value for this coefficient, which does not establish priority. This note provides an explicit certified evaluation and closes the finite condition in P2588; it does not claim a new general theorem or a new asymptotic 2-SAT threshold. The primary journal article is article 7, correcting article 8 in the supplied problem metadata.

AI assistance. ChatGPT (OpenAI) was used to derive and implement the exact recurrences, design the independent verification, execute computations, audit sources and draft this note. The mathematical argument and computational claims should be assessed from the supplied code and certificates. The exact scope of the Lean checks is stated in Section 5. This version records no journal submission.

References

[1] S. Dovgal, E. de Panafieu and V. Ravelomanana. *Exact enumeration of satisfiable 2-SAT formulae*. Combinatorial Theory **3**(2) (2023), article 7. doi:10.5070/C63261985. arXiv:2108.08067v2.

[2] S. Dovgal and E. de Panafieu. Companion repository, *Enumeration-2sat-aux*, commit `346079fa8606ebd48f4557bcef7bc1f73a2aed4d`. https://gitlab.com/sergey-dovgal/enumeration-2sat-aux.

[3] TheoremDB contributors. P2588, *Median satisfiability threshold for a six-variable clause set*. https://theoremdb.org/statements/P2588/. Accessed 20 October 2026.